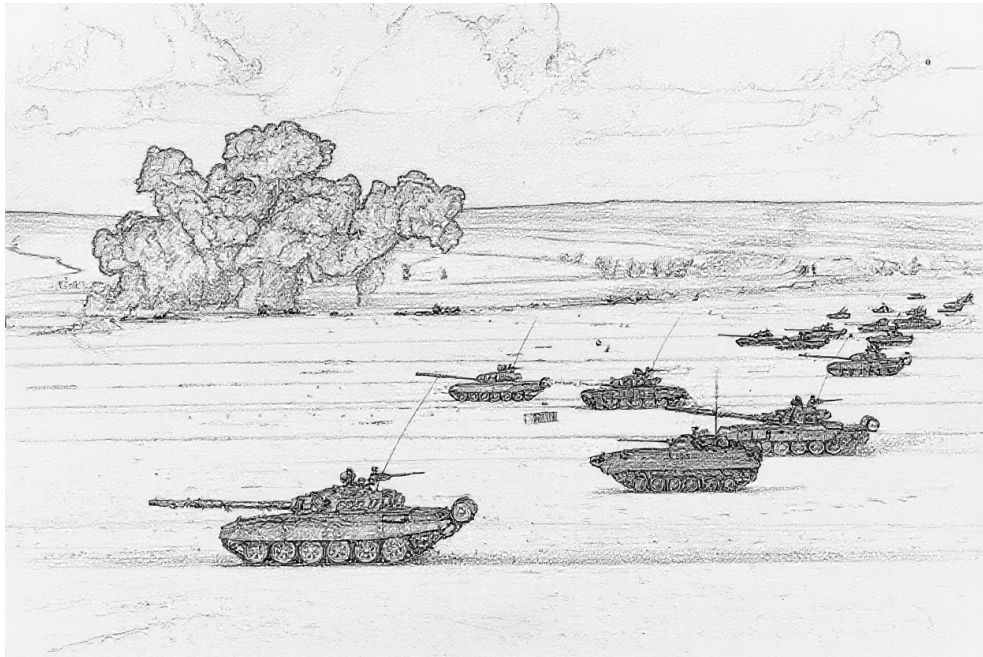


Air Power

Ground Unit Briefing

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INTRODUCTION

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:¹

“You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb.”

This briefing document deals with the ground units and ground targets that are subject to air-to-ground attacks. It first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from *Air Power Journal*, and from Malcolm Pipe’s extensive compendium of AAA and SAM units. However, the data have been carefully revised to improve their accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded. Extensive notes accompany the main text, providing provenance of data, sources of information, and justifications for decisions.

I am grateful to Carl Smeaton for many discussions on the data presented here and to Malcolm Pipes for leading the way with his compilations of AAA and SAM units.

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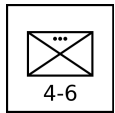
GROUND-COMBAT AND TRANSPORT UNITS

Ground-combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide operational and strategic mobility to infantry and towed artillery and are the lifeblood of logistic networks. These units are presented below with their counters. The Ground-Combat and Transport Unit Table summarizes their properties. Some representative examples of vehicles are given in the descriptions, and the Vehicle Types Table gives a more extensive list. All ground-combat and transport platoons and batteries are also available as sections and squads.²

INFANTRY UNITS

Infantry are soldiers trained to fight on foot and, in some cases, from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest.

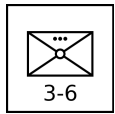
An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.



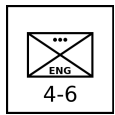
Infantry Platoon: An infantry platoon is armed with small arms, light machine guns, and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. The platoon is the primary combat unit in larger infantry formations.



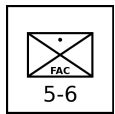
Infantry Weapons Platoon: An infantry weapons platoon is equipped with heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers. It is often a company- or battalion-level asset, and its role is to provide direct-fire support to the infantry platoons in its formation.³



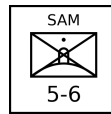
Infantry Mortar Platoon: An infantry mortar platoon is equipped with 60-mm, 81-mm, 82-mm, or similar mortars. It is often a battalion-level asset, and its role is to provide indirect fire support and smokescreens to the infantry platoons in its formation. Due to the weight of ammunition required for sustained fire, it will often be transported by a light truck platoon even in light infantry formations.



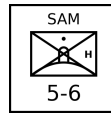
Infantry Engineers Platoon: An infantry engineers platoon is additionally equipped and trained for combat engineering tasks such as mine-laying and mine-clearing. It is often battalion-level asset.



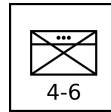
Infantry FAC Squad: An infantry FAC squad has a forward air controller with other soldiers providing communication and protection. An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry SAM Squad: Infantry SAM squads are equipped with shoulder-launched SAMs, with one launcher per squad. Their SAM capabilities and VP values are given elsewhere. An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry Heavy SAM Squad: Infantry heavy SAM squads are equipped with pedestal-mounted SAMs, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section. An infantry heavy SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.

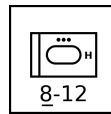


Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. It consists of the formation commander, their staff, and their protection detail.⁴

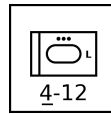
ARMORED VEHICLE UNITS

Armored ground-combat vehicle units combine armored protection for their crew and weapons, mobility, and either a fire-power or transport capability.

The primary factors that determine an armored-vehicle unit's type are its weapons and protection level. The means of traction is a secondary consideration. Armored cars and armored half-tracks are considered to be light tanks, light IFVs, or light or open APCs, light anti-tank missile vehicles, or light reconnaissance vehicles according to their role and armament.



Tank Platoon: A tank platoon has armored vehicles principally armed with a direct-fire gun. The gun is most commonly mounted in a turret, but AFVs with casemate guns such as the SU-100, Kanonen JPz, and Strv 103 are also considered to be tanks. The role of a tank is the support of infantry and the destruction of other tanks and armored vehicles. A tank platoon may have heavy, medium, light, or open armor.⁵

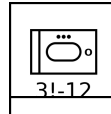


Heavy examples: M1 Abrams, T-64, and T-72/80.

Medium examples: M60, T-34, T-54/55, and T-62.

Light examples: M114A2, M551 Sheridan, and PT-76.

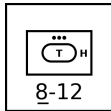
Open example: ASU-57.



IFV Platoon: An infantry fighting vehicle (IFV) platoon is equipped with armored vehicles that provide infantry with tactical and operational mobility, protection, and fire support, being armed with a 20 mm or larger cannon and sometimes an anti-tank missile launcher. In most cases, the infantry must dismount to fight effectively. An IFV platoon may have medium or light armor.⁶

Medium example: M2A2 Bradley.

Light examples: M2/M2A1 Bradley and BMP-1/2.



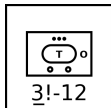
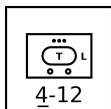
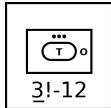
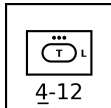
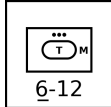
APC Platoon: An APC platoon is equipped with armored vehicles that provide infantry with tactical and operational transport and protection, but unlike IFVs do not provide fire support and at most are armed with a heavy machine gun. An APC platoon may have heavy, medium, light, or open armor. Heavy and medium APCs are typically converted tanks.⁷

Heavy example: Nakpadon.

Medium example: Nagmachon.

Light examples: M113 and BTR-50PK/60PB/70.

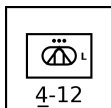
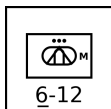
Open examples: M3 half-track and BTR-50P/60P.



Armored Truck Platoon: An armored truck platoon has armored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition. Unlike a truck platoon, it has armored protection, but unlike an APC platoon, it has more limited off-road mobility. An armored truck platoon may have light or open armor.⁸

Light example: BTR-152K.

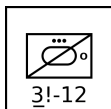
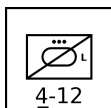
Open examples: BTR-40 and other BTR-152 versions.



Armored Anti-Tank Missile Platoon: An armored anti-tank missile platoon has armored vehicles equipped with anti-tank guided missiles. Its mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers such as the Kanonen JPz are considered to be tanks. An armored anti-tank missile platoon is often a battalion- or regiment-level asset. It may have medium or light armor.⁹

Medium examples: Rakaten JPz and Jaguar.

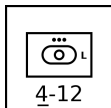
Light examples: M150, M901, and BRDM.



Armored Reconnaissance Platoon: An armored reconnaissance platoon has armored vehicles specialized in scouting and observation and with no offensive weapons more powerful than machine guns. (Vehicles used for reconnaissance that do have significant offensive weapons are typically classified as light tanks or IFVs.) An armored reconnaissance platoon may have light or open armor.¹⁰

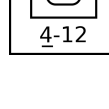
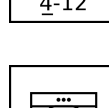
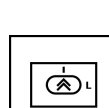
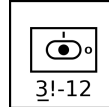
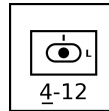
Light examples: M114 and BRDM-2.

Open example: M20.



Armored Mortar Platoon: An armored mortar platoon has mortars mounted in armored vehicles (most commonly converted APCs or IFVs). It is typically a battalion-level asset in mechanized-infantry or tank battalions, and its role is indirect fire support. An armored mortar platoon has light armor.¹¹

Examples: M106 and M125.



Armored Artillery Battery: An armored artillery battery has guns or howitzers for indirect-fire support in armored vehicles providing some protection, in particular from counter-battery fire, and mobility. It is typically a regiment-, brigade-, or division-level asset. An armored artillery battery may have light or open armor.¹²

Light examples: M109, 2S3, and 2S1.

Open example: M7 Priest.

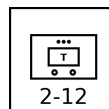
Armored Rocket Artillery Battery: An armored rocket artillery battery is the rocket-armed equivalent of an armored artillery battery. An armored rocket artillery battery has light armor.¹³

Example: M270.

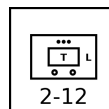
Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized-infantry, or armored-artillery battalion, regiment, or brigade. It consists of the formation commander, their staff, and their protection detail. It is typically equipped with modified APCs or IFVs. An armored HQ platoon has light armor.

UNARMORED VEHICLE UNITS

Unarmored or soft vehicles provide mobility, but do not provide (or provide only incomplete) armor protection for their crew and weapons. They include armored vehicles whose crew or weapons are exposed. They may be wheeled or tracked.

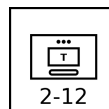


Truck Platoon: A truck platoon has unarmored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition. It can provide operational mobility to infantry and towed artillery, but due to its extreme vulnerability to direct- and indirect-fire, is not typically used in a tactical role.



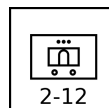
Light Truck Platoon: A light truck platoon has unarmored wheeled vehicles that can transport light loads of infantry, towed weapons, and supplies. It can provide transport to independent infantry FAC or SAM units, but in these roles it can avoid close contact with enemy combat units, mitigating its vulnerability.

Examples: M151 Jeep, HMMWV, and GAZ-69.



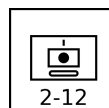
Tracked Carrier Platoon: A tracked carrier platoon is equipped with unarmored tracked vehicles that can transport medium loads of infantry, towed weapons, and supplies, including fuel and ammunition. It serves the same roles as trucks or light trucks, but in terrain that requires their greater mobility, in particular arctic regions.

Example: Bv 202.



Mobile Anti-Tank Missile Platoon: A mobile anti-tank missile platoon is equipped with anti-tank missiles mounted on unarmored light trucks. It is extremely vulnerable to direct- and indirect-fire, and must rely on concealment and mobility.

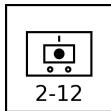
Examples: M151 TOW and M966 HMMWV TOW.



Tracked Artillery Battery: A tracked artillery battery has guns or howitzers mounted on tracked vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its

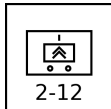
role is indirect fire support.¹⁴

Examples: M107, M110, 2S4, and 2S7.



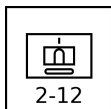
Mobile Artillery Battery: A mobile artillery battery has howitzers mounted on wheeled vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. Lacking armor, it must avoid counter-battery fire by using their mobility to rapidly change firing positions. It is typically a regiment-, brigade-, or division-level asset, and their role is indirect fire support.¹⁵

Examples: CAESAR and Archer.



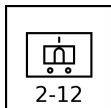
Mobile Rocket Artillery Battery: A mobile rocket artillery battery is the rocket-armed equivalent of a mobile artillery battery. Again, it relies on mobility to avoid counter-battery fire.¹⁶

Examples: M142 HIMARS and BM-21 Grad.



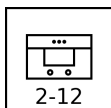
Tracked Missile Artillery Battery: A tracked rocket artillery battery is the rocket-armed equivalent of a tracked artillery battery. The tracked chassis provides mobility, but the missiles are exposed. Nevertheless, the long range of the missile permits deployment far behind the front line, mitigating its vulnerability.¹⁷

Example: M752 Lance.



Mobile Missile Artillery Battery: A mobile missile artillery battery is the missile-armed equivalent of a mobile artillery battery. Again, the long range of the missile permits deployment far behind the front line, mitigating its vulnerability.¹⁸

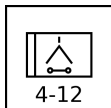
Examples: M1003 Pershing II and OTR-21 Tocha (SS-21).



Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion, regiment, or brigade other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.

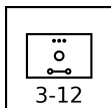
TOWED UNITS

Towed weapons provides neither inherent mobility nor protection. However, they are light and cheap compared to similar vehicle-mounted weapons. Towed weapons are typically provided with appropriate ground or helicopter transport.



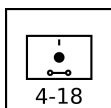
Towed Anti-Tank Gun Battery: A towed anti-tank gun battery has anti-tank guns on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Example: 2A18 (T-12) and 2A45 (Sprut).



Towed Mortar Platoon: A towed mortar platoon has 120 to 160 mm mortars on unprotected, towed mounts. It is often a regimental- or brigade- asset. Smaller infantry mortars are found in infantry mortar platoons.

Example: M1943.



Towed Artillery Battery: A towed artillery battery has guns or howitzers on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Examples: M119, M198, and 2A18 (D-30).

GROUND-COMBAT AND TRANSPORT UNIT TABLE

Type	Size ^a	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class ^b
Infantry	Platoon	4	6	U	5/3/2	B2
Infantry Weapons	Platoon	3	6	U	6/4/2	B2
Infantry Mortar	Platoon	3	6	U	6/4/2	B2
Infantry FAC	Squad	5	6	U	-/-/8	—
Infantry SAM	Squad	5	6	U	—	—
Infantry Heavy SAM	Squad	5	6	U	—	—
Infantry HQ	Platoon	4	6	U	10/7/3	B2
Heavy Tank	Platoon	8	12	U	12/8/4	B3
Medium Tank	Platoon	6	12	U	10/7/3	B3
Light Tank	Platoon	4	12	U	6/4/2	B3
Open Tank	Platoon	3!	12	U	5/3/2	B3
Medium IFV	Platoon	6	12	U	10/7/3	B3
Light IFV	Platoon	4	12	U	6/4/2	B3
Heavy APC	Platoon	8	12	U	10/7/3	B3
Medium APC	Platoon	6	12	U	8/5/3	B3
Light APC	Platoon	4	12	U	6/4/2	B3
Open APC	Platoon	3!	12	U	5/3/2	B3
Light Armored Truck	Platoon	4	12	G/P	6/4/2	B3
Open Armored Truck	Platoon	3!	12	G/P	5/3/2	B3
Medium Armored Anti-Tank Missile	Platoon	6	12	U	12/8/4	B3
Light Armored Anti-Tank Missile	Platoon	4	12	U	10/7/3	B3
Light Armored Reconnaissance	Platoon	4	12	U	6/4/2	B3
Open Armored Reconnaissance	Platoon	3!	12	U	4/3/1	B3
Light Armored Mortar	Platoon	4	12	U	10/7/3	B3
Light Armored Artillery	Battery	4	12	U	12/8/4	B3
Open Armored Artillery	Battery	3!	12	U	12/8/4	B3
Light Armored Rocket Artillery	Battery	4	12	U	12/8/4	B3
Light Armored HQ	Platoon	4	12	U	12/8/4	B3
Truck	Platoon	2	12	G/P	4/3/1	—
Light Truck	Platoon	2	12	G/P	4/3/1	—
Tracked Carrier	Platoon	2	12	U	4/3/1	—
Mobile Anti-Tank Missile	Platoon	2	12	G/P	8/5/3	—
Mobile Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Artillery	Battery	2	12	U	10/7/3	—
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Missile Artillery	Battery	2	12	U	10/7/3	—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3	—
Mobile HQ	Platoon	2	12	G/P	10/7/3	—
Towed Mortar	Platoon	3	12	T	7/5/2	—
Towed Anti-Tank Gun	Battery	4	12	T	8/5/3	—
Towed Artillery	Battery	4	18	T	8/5/3	—

^a Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences and VPs. ^b Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

VEHICLE TYPES TABLE

Heavy Tanks		Light Tanks		Heavy APCs		Medium Armored Anti-Tank		Light Truck				
Al-Khalid/VT-1A	PK/CN 2001	AMX-10 RC	FR 1981	Achzarit	IL 1988	Raketen JPz 1	DE 1961	GAZ-69	SU 1953			
Ariete	IT 1995	AMX-13	FR 1953	Nakpadon	IL 1998	Raketen JPz 2	DE 1967	HMMWV	US 1985			
Arjun	IN 2004	Fox	GB 1975	Namer	IL 2008	Jaguar 1	DE 1978	Ilitis	GE 1978			
Challenger 1	GB 1983	Commando (90 mm)	US 1964	Puma	IL 1993	Jaguar 2	DE 1983	Landrover	UK 1949			
Challenger 2	GB 1998	LAV I Cougar	CA 1976	Medium APCs		Light Armored Anti-Tank		M38 1/4-ton truck	US 1949			
Chieftain	GB 1967	LAV II Coyote	CA 1990	Boxer APC	NL/DE 2011	BRDM-2 ATGM	SU 1973	M151 1/4-ton truck	US 1960			
Conqueror	GB 1955	M8/Greyhound	US 1943	Nagmachon	IL 2000	BRDM-3 ATGM	SU 1973	UAZ-469	SU 1971			
IS-2	SU 1944	M24 Chaffee	US 1944	Nagmashot	IL 1990	Fennek MRAT	DE/NL 2003	Willys MB 1/4-ton truck	US 1941			
IS-3	SU 1945	M41 Walker Bulldog	US 1953	Light APCs		FV438 Swingfire	GB 1971	Tracked Carrier				
K1	KR 1987	M113 C&V with 25 mm	US 1974	Bronco/Warthog	SG 2001	M150	US 1973	AT-T/S/L	SU 1950			
K2 Black Panther	KR 2014	M114A2	US 1969	BvS10/Viking	SE 2005	M901 ITV	US 1979	Bv 202	SE 1964			
Leclerc	FR 1992	M551 Sheridan	US 1969	BTR-50PK	SU 1958	Striker	GB 1976	Bv 206	SE 1980			
Leopard 2	DE 1979	Panhard AML	FR 1961	BTR-60PA/PAI/PB	SU 1965	Stryker ATGM	US/CA 2003	BvS10/Beowulf	SE 2024			
M-84	YU 1985	Panhard EBR	FR 1951	BTR-70	SU 1972	VAB HOT	FR 1978	M548	US 1960			
M1 Abrams	US 1980	PT-76	SU 1951	BTR-80	SU 1986	YP-408 PRAT	NL 1976	Sisu Nasu	FI 1986			
Merkava 1	IL 1979	Saladin	GB 1958	Commando	US 1963	YPR-765 PRAT	NL 1980	Mobile Anti-Tank Missile				
Merkava 2	IL 1983	Scimitar	GB 1973	FV432	GB 1963	Light Armored Reconnaissance		M151 TOW	US 1970			
Merkava 3	IL 1989	Scorpion	GB 1973	LAV I Grizzly	CA 1976	BRDM-2	SU 1973	M966 HMMWV TOW	US 1985			
Merkava 4	IL 2004	SpPz Luchs	DE 1975	LAV II Bison	CA 1990	Fennek	DE/NL 2003	Mobile Artillery				
PT-91	PL 1995	Stryker MGS	US/CA 2006	LVTP-5	US 1956	Ferret	GB 1952	Archer	SE 2016			
Stridsvagn 103	SE 1967	Strv 74	SE 1958	LVTP-7 (AAV)	US 1972	M113 C&V/Lynx	US 1966	CAESAR	FR 2004			
T-10	SU 1954	Type 62	CN 1961	M59	US 1954	M114	US 1962	Tracked Artillery				
T-14 Armata	RU 2024	Type 63	CN 1963	M75	US 1952	Open Armored Reconnaissance		2S4 Tyulpan	SU 1975			
T-64	SU 1966	Type 05 AAV	CN 2005	M113	US 1960	M20	US 1943	2S7 Pion	SU 1976			
T-72	SU 1973	Type 08 ARV	CN 2008	OT-64 SKOT	CS/PL 1963	Light Armored Mortar		M41	US 1942			
T-80	SU 1976	Open Tanks		OT-810 half-track	CS 1958	FV432 with 81 mm	GB 1963	M107	US 1962			
T-84	UA 1999	ASU-57	SU 1951	Piranha APC	CH 1974	M106	US 1964	M110	US 1963			
T-90	RU 1992	SU-76	SU 1943	Saracen	GB 1952	M125	US 1964	AMX-13 F3	FR 1962			
Type 10	JP 2012	Medium IFVs		Spartan	GB 1978	M1064	US 1990	Mobile Rocket Artillery				
Type 90	JP 1990	BMP-3	SU 1987	Stryker ICV	US/CA 2002	Stryker MCV	US/CA 2005	BM-8	SU 1941			
Type 99	CN 2001	Boxer IFV	NL/DE 2011	TPz Fuchs	DE 1979	Light Armored Artillery		BM-13	SU 1941			
Medium Tanks		Strf 9040/CV90	SE 1994	Type 63	CN 1978	2S1 Gvozдика	SU 1972	BM-31	SU 1944			
AMX-30	FR 1966	LAV III (add-on-armor)	CA 2009	Type 77	CN 1978	2S3 Akatsiya	SU 1971	BM-24	SU 1947			
ASU-85	SU 1959	M2A2/3/4 Bradley	US 1988	Type 89	CN 1978	2S19 Msta-S	SU 1989	BM-14	SU 1952			
Centurion	GB 1945	M3A2/3/4 Bradley	US 1988	Type 08 APC	CN 2008	AMX-30 AuF1	FR 1977	BM-21 Grad	SU 1963			
Comet	GB 1944	SPz Marder	DE 1971	VAB	FR 1979	AS-90	GB 1992	BM-27 Uragan	SU 1975			
Cromwell	GB 1944	SPz Puma	DE 2015	YP-408	NL 1964	Bkan 1	SE 1967	BM-30 Smerch	SU 1989			
ISU-122	SU 1944	Light IFVs		Open APCs		L-33 Ro'Em	IL 1973	M142 HIMARS	US 2005			
ISU-152	SU 1943	AIFV	US 1979	AT-P	SU 1954	M52	US 1955	Tracked Missile Artillery				
Kanonen JPz	DE 1965	AMX-10P	FR 1973	BTR-50P	SU 1952	M108	US 1962	M752 Lance	US 1972			
Leopard 1	DE 1965	BMP-1	SU 1966	BTR-60P	SU 1959	M109	US 1963	AMX-30 Pluton	FR 1974			
M-50 Sherman	IL/FR 1956	BMP-2	SU 1980	M3 half-track	US 1941	Panzerhaubitze 2000	DE 1998	Mobile Missile Artillery				
M-51 Sherman	IL/FR 1961	Commando (20 mm)	US 1963	Light Armored Trucks		Open Armored Artillery		9K52 Luna-M (Frog-7)	SU 1962			
M103	US 1957	FV432 RARDEN	GB 1975	BTR-152K	SU 1959	AMX-105A	FR 1958	Corporal	US 1954			
M26 Pershing	US 1944	LAV III	CA 1999	Saxon	GB 1983	M-50	IL 1962	Sergeant	US 1962			
M4 Sherman	US 1942	LAV-25	CA 1983	Open Armored Trucks		M7 Priest	US 1942	M34/M50 Honest John	US 1953			
M46 Patton	US 1949	M2 and M2A1 Bradley	US 1981	BTR-40	SU 1950	M40	US 1945	M74 Redstone	US 1958			
M47 Patton	US 1951	M3 and M3A1 Bradley	US 1981	BTR-152A	SU 1950	M44	US 1954	M790 Pershing 1a	US 1962			
M48 Patton	US 1952	M113A1 FSC	AU 1972	SKP/VKP	SE 1944	Light Armored Rocket Artillery		M1003 Pershing II	US 1983			
M60	US 1959	Piranha IFV	CH 1974	SKPF/VKPF	SE 1956	M270 MLRS	US 1983	OTR-21 Tochka (SS-21)	SU 1975			
Panzer 61	CH 1965	Pbv 301	SE 1961	Open APCs		BM-1	RU 1988	RSD-10 Pioneer (SS-20)	SU 1976			
Panzer 68	CH 1971	Pbv 302	SE 1966	BTR-40	SU 1950							
SU-100	SU 1944	SPz Kurz	FR 1959	BTR-152A	SU 1950							
T-34	SU 1940	SPz Lang HS.30	CH 1960	SKP/VKP	SE 1944							
T-54	SU 1955	Stryker ICVD	US 2018									
T-55	SU 1958	Type 86	CN 1992									
T-62	SU 1961	Type 92	CN 1992									
TAM	AR 1983	Type 05 IFV	CN 2005									
Type 59	CN 1959	Type 08 IFV	CN 2008									
Type 61	JP 1961	VCBI	FR 2008									
Type 69	CN 1972	Warrior	GB 1987									
Type 74	JP 1975	YPR-765	NL 1977									
Type 79	CN 1984											
Type 85	CN 1988											
Type 88	CN 1988											
Type 96	CN 1997											
Type 15	CN 2018											
Vickers MBT	GB 1965											
Vijayanta	IN/GB 1965											

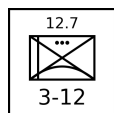
AAA UNITS

AAA units use guns to destroy attacking aircraft, but also have a secondary ground-combat capability. Some also have SAMs, giving a layered capacity. AAA units are presented below with their counters, ordered first by caliber and then by year of introduction.

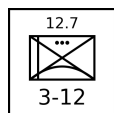
The AAA Unit Table summarizes the properties of AAA units. As well as the basic properties common to all ground units, the table gives: the AAA class (light, medium, or heavy); the maximum altitude; the limits of short, medium, and long range in hexes; the hit rolls at short, medium, and long range; the damage rating; whether the unit has an all-weather or ranging-only fire-control radar and its frequency band or a laser rangefinder; whether it has night infrared sights; and whether it is also equipped with a SAM system. If the unit has a SAM, this will be described in detail in the section on SAM launcher units.¹⁹

All AAA platoons and batteries are also available as sections (but not squads), with damage resiliences of 2, hit rolls reduced by 1, and VPs corresponding to a platoon or battery that has taken 1D damage.

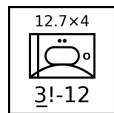
LIGHT AAA UNITS



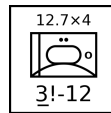
M2 Platoon: The Browning M2 .50 cal (12.7 mm) heavy machine gun is normally found on a low tripod for ground combat or mounted on armored fighting vehicles, but here is used on a tall dual-purpose mount suitable for air defense. A platoon has six guns. The M2 gun was introduced in 1933 and has been used by United States forces and their allies in numerous conflicts since then.²⁰



DShK-38 Platoon: The DShK-38 12.7 mm heavy machine gun is most commonly found on a low, wheeled mount for ground combat or mounted on armored fighting vehicles, but here is used on a tall tripod suitable for air defense. It is nicknamed the Dushka (darling) or Dochka (daughter). A platoon has six guns. The DShK-38 entered service with the Soviet Army in 1938 and saw action in World War II and numerous conflicts since then, including the Korean and Vietnam Wars, in which the Soviet Union participated or provided arms. In particular, it was the main air-defense gun for Chinese and North Korean field units in the early part of the Korean War.²¹

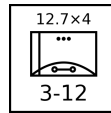


M16 Section: The M16 Multiple Gun Motor Carriage has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a modified M3 half-track. A section has two vehicles. The M16 is a development on the M13 and M14 Multiple Gun Motor Carriages, which have two M2 heavy machine guns in a similar turret. As a single M2 has an insufficient rate of fire to be effective even against WW2 aircraft, first two and then four were used on a single mount. The M16 was used by United States forces during World War II. It was also used by US, Commonwealth, and South Korean forces in the Korean War, primarily in an anti-infantry role. It was largely withdrawn from US services in late 1951 and officially retired in 1958. Israeli forces employed it during the 1956 Arab-Israeli

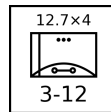


War. Other post-WW2 users include Belgium, France, West Germany, Japan, Netherlands, Philippines, Portugal, and Thailand.²²

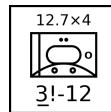
M17 Section: The M17 Multiple Gun Motor Carriage is the equivalent of the M16 Multiple Gun Motor Carriage, but based on the slightly different M5 half-track. A section has two vehicles. The M17 was supplied to the Soviet Union and entered service in 1944. It was presumably replaced by the BTR-40A.²³



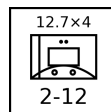
M55 Platoon: The M55 has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns (the same turret as used in the M16) mounted on a towed carriage. A platoon has four gun mounts. The M55 was used by United States forces during the Korean War and by Pakistani forces during the 1965 India-Pakistan War.²⁴



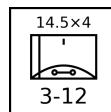
Vz.53 Platoon: The Czechoslovak Vz.53 has four Vz.38/46 12.7 mm machine guns (copies of the Soviet DShK) on a light, wheeled mount. It is sometimes referred to as the M53. A platoon has six guns. It entered service with the Czechoslovak Army in 1953, and was exported to Cuba, Egypt (from 1955), Iraq, Syria, and Vietnam. Cuba subsequently exported some to Grenada. It saw action with the Egyptian Army in the 1956 War, with Cuba in the Bay of Pigs Invasion in 1961, in the Vietnam War, in the Nicaraguan Revolution, with the Grenadian Army during the invasion by US force in 1983.²⁵



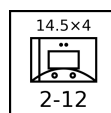
BTR-152A Vz.53 Section: The BTR-152A Vz.53 has a Czechoslovak Vz.53 gun mount with four 12.7 mm Vz.38/46 machine guns in the open, six-wheeled Soviet BTR-152 APC. The gun mount is also referred to as the M53. A section has two vehicles. It served only in the Egyptian Army, and saw action in the 1956 War.²⁶



Mobile M45 Section: The mounted M45 has the M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns (the same turret as used in the M16) removed from an M55 trailer and mounted on a truck. During the Vietnam War, in 1967, US forces improvised these *gun trucks* for convoy security.²⁷



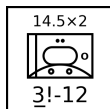
ZPU-1/2/4 Battery: ZPU-1 battery ZPU-2 battery The ZPU-1/2/4 have one, two, or four 14.5 mm heavy machine guns in open mounts on towed carriages. A battery has six gun mounts. The ZPU-1/2/4 was introduced in 1949 and saw extensive service with Soviet or Soviet-supported forces in many subsequent conflicts, including with communist forces in the Korean War and Vietnam War and with Arab forces in the various Arab-Israeli Wars from 1956 onwards.²⁸



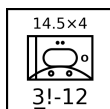
Mobile ZPU-1/2/4 Section: mobile ZPU-1 section mobile ZPU-2 section

The mobile ZPU-1/2/4 is a light truck with a ZPU-1/2/4 gun mount. A section has two vehicles. Many

users of the ZPU-1/2/4 adapt it in this way for increased mobility, and like the towed versions, these mounted versions have seen combat in many conflicts. Formal conversions tend to be used for air-defense, but informal conversions or *technicals* are mainly used for ground combat, serving as light cavalry or infantry support.²⁹

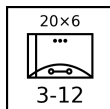


BTR-40A Section: The BTR-40A has a twin 14.5 mm ZPU-2 turret in the open, four-wheeled BTR-40 APC. A section has two vehicles. The BTR-40A entered service with the Soviet Army in 1950, presumably replacing the M17. It was also used by East Germany and Cuba.³⁰

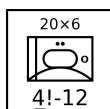


BTR-152A ZPU-2/4 Section: BTR-152A ZPU-2 section

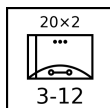
The BTR-152A has a twin 14.5 mm ZPU-2 or quad 14.5 mm ZPU-4 turret in the open, six-wheeled BTR-152 APC. A section has two vehicles. The BTR-152A entered service with the Soviet Army in 1951, presumably replacing the BTR-40A. The BTR-152D/E are similar, but based on the improved BTR-152V/V1 variants. Other users include Ethiopia in the Ogaden War and Egypt.³¹



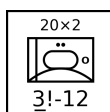
M167 Platoon: The M167 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in an open mount on a towed carriage. A platoon has four gun mounts. The fire-control system has optical sights and a range-only radar. The M167 was introduced into service with the US Army in 1967, principally for airfield defense, and subsequently served with other US allies.³²



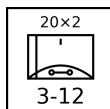
M163 Section: The M163 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in a lightly armored but open-topped turret on an adapted M113 APC. A section has two vehicles. The fire-control system has optical sights and a range-only radar. The M163 VADS was introduced into service with the US Army in 1968 and has since been adopted by other US allies. The M163 was used by US forces in the Vietnam War, the 1989 Invasion of Panama, and the 1991 Gulf War. It was also used by the Israel Army in the 1982 Lebanon War and in 2002 during the Second Intifada. From 1988 in the US Army, each M163 was typically issued with a FIM-92 Stinger launcher with two rounds, so each M163 section effectively carried a pair of FIM-92 squads.³³



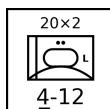
TCM-20 Platoon: The TCM-20 is an Israeli adaptation of the M55, with two Hispano-Suiza 20mm guns from obsolete aircraft replacing the four .50 cal machine guns. As with the M55, the fire-control system is entirely optical. A platoon has four gun mounts. The TCM-20 was employed by Israeli forces from 1970 and subsequently saw combat in the 1973 Arab-Israeli War and the 1982 Lebanon War.³⁴



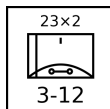
M3 TCM-20 Section: The M3 TCM-20 has a TCM-20 turret Hispano-Suiza 20mm guns mounted in an adapted M3 half-track, like the earlier M16. A section has two vehicles. The M3 TCM-20 was used by Israeli forces in the 1973 Arab-Israeli War and the 1982 Lebanon War.³⁵



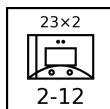
Twin Rh-202 Battery: The Rheinmetall Rh-202 has a single or twin 20mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. The most common version is the twin mount. A battery has six gun mounts. The twin version entered service with West German forces in 1972 and was used for airfield defense. Subsequently, it was exported and notably used by the Argentine Air Force in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).³⁶



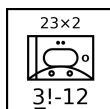
Panhard M3 VDA Section: The Panhard M3 VDA vehicles have an armored turret for twin HS.820 20 mm guns mounted on the Panhard M3 APC. A section has two vehicles. The guns have optical sights, but typically one vehicle in each section or platoon is equipped with a search-and-ranging radar and can share this information automatically with the others. The Panhard M3 VDA served with the armed forces of Côte d'Ivoire, Niger, and the UAE.³⁷



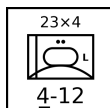
ZU-23 Battery: The ZU-23 has twin 23 mm cannons in an open mount on a towed carriage. It was designed as a more powerful replacement for the ZPU-1/2/4. A battery has six gun mounts. The fire-control system is entirely optical. The ZU-23 was introduced into service by the Soviet Army in 1960. It later served with the armed forces of many Soviet allies, including with North Vietnam in the Vietnam War, with Egypt and Syria in the 1967 and 1973 Wars, with Soviet forces in the Soviet-Afghan War, with Iraq in the Iraq-Iran War, with Syria in the 1982 Lebanon War, and with both sides in the Afghan Civil War.³⁸



Mobile ZU-23 Section: This is a section of two light trucks mounting ZU-23 guns. As with the earlier ZPU-1/2/4, many users of the ZU-23 mount it on vehicles, including trucks, tracked carriers, and APCs, for better mobility. Informal conversions are typical used as *technicals* in ground combat.³⁹

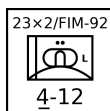


BTR-152A ZU-23 Section: The BTR-152A ZU-23 has a twin 23 mm ZU-23 turret in the open, six-wheeled BTR-152 APC. A section has two vehicles. It entered service with the PLO prior to the 1982 War in Lebanon; the in-service data is a guess.⁴⁰



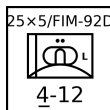
ZSU-23-4 Section: The ZSU-23-4 has four radar-controlled 23 mm guns, similar to those in the ZU-23, mounted in an armored turret on a hull adapted from the PT-76 light tank. A section has two vehicles. The ZSU-23-4 entered service in the Soviet Army in 1965, replacing the ZSU-57-2 and outclassing all NATO anti-aircraft guns at that time. A platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments and later complemented by a platoon of four SA-9 vehicles. It was exported to Soviet al-

lies and saw extensive combat with Arab forces in the 1973 Arab-Israeli War and with Iraqi forces in the Iran-Iraq War and the 1991 Gulf War.⁴¹

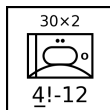


Sinai 23 Section: Sinai 23 Ayn-al-Saqr section

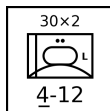
The Sinai 23 has a Dassault TA20 turret with two 23 mm guns and six Ayn-al-Saqr (similar to the Strela-2M) or Stinger missiles on an M113 chassis. A section has two vehicles. Each battery of two sections operates with one vehicle equipped with an RA 20S search and ranging radar. The Sinai 23 entered service with the Egyptian Army in 1991.⁴²



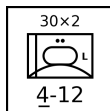
LAV-AD Section: The LAV-AD has a 25 mm GAU-12 rotary cannon and eight FIM-92D Stinger missiles mounted in a turret on the LAV-25 vehicle. A section has two vehicles. The LAV-AD does not have radar, but is equipped with a passive night IR sight. It entered service with the USMC in 1997, but was retired after only a few years.⁴³



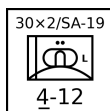
M53/59 Section: The M53/59 has a twin M53 30 mm gun mount on a partially armored truck. The fire-control system is entirely optical. A section has two vehicles. It entered service in the Czech Army in 1959 and was used instead of the ZSU-57-2. It was also exported to Iraq and saw combat in the Iran-Iraq War and the 1991 Gulf War.⁴⁴



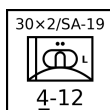
AMX-13DCA Section: The AMX-13DCA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-12 tank. A section has two vehicles. The AMX-13DCA is equipped with the Thomson-CSF Oeil Noir radar. It served only in the French Army, entering service in 1969 and being retired in the 1990s.⁴⁵



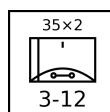
AMX-30SA Section: The AMX-30SA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-30 tank. A section has two vehicles. AMX-30SA is an improved version of the AMX-30 DCA and served only in the Saudi Army from 1979.⁴⁶



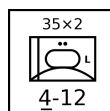
Tunguska Section: The Tunguska has twin radar-controlled 30mm guns and four SA-19 missiles in an armored turret on an armored and tracked hull. A section has two vehicles. The Tunguska was designed to counter the A-10 and AH-64, against which the earlier ZSU-23-4 had weaknesses. It entered service in the Soviet Army in 1984. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported. The Tunguska is referred to as the "ZSU-30-2" in *Air Strike*.⁴⁷



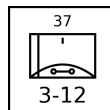
Tunguska-M Section: The Tunguska-M is an improved version of the Tunguska with eight missiles instead of four. A section has two vehicles. It entered service in the Soviet Army in 1990. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.⁴⁸



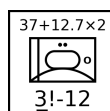
Oerlikon GDF Battery: The Oerlikon GDF has two 35 mm guns in an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but a section is often coupled with the Super Fledermaus or Skyguard fire-control radars. (These radars cannot control a whole battery, only a section.) The Oerlikon GDF entered service in 1963 and has served widely. Notably, it saw combat in the South Atlantic War with the Argentine forces in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).⁴⁹



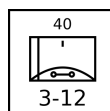
Gepard Section: The Gepard has two radar-controlled 35 mm guns in an armored turret on an adapted Leopard 1 tank hull. A section has two vehicles. The Gepard entered service in 1976 with the West German Army and also served with the Belgian and Dutch armies. After the end of the Cold War, they were retired by these armies, and many were sold on to other countries.⁵⁰



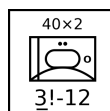
61-K Battery: The 61-K (M1939) has a single 37 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system is entirely optical, and (as an exception to the normal rule allowing medium AAA units to employ external FCRs) the 61-K cannot be coupled to an external fire-control radar. After entering service in 1939 with the Soviet Army, it served in WW2 and after until replaced by the S-60 57 mm gun. It also served with many Soviet allies, and in particular with communist forces in the Korean and Vietnam Wars. The 61-K is referred to as the "M-38" in *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*.⁵¹



M15 Section: The M15 Combination Gun Motor Carriage has a M1 37-mm cannon and two Browning M2 .50 cal (12.7 mm) machine guns in an M42 turret mounted on a modified M3 half-track. A section has two vehicles. The M15 entered service with the US Army in 1942, saw combat in WW2 and the Korean War, in the latter mainly in a ground-support role, and was retired in 1953. It also served with the armed forces of Japan and Yugoslavia.⁵²



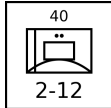
Bofors L/60 Battery: The Bofors L/60 has a single 40 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system is entirely optical. The Bofors L/60 entered service in the 1940s and was widely used by Allied forces during WW2, both on land and sea. After the war, many continued to see service, although it was increasingly inadequate against faster jet aircraft and in many cases was replaced by the new L/70 model.⁵³



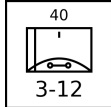
M19 Section: The M19 Multiple Gun Motor Carriage has two Bofors 40 mm L/60 guns in a partially armored mount on a tracked chassis based on that of the M24 light tank. A section has two vehicles. The fire-control system is entirely optical. The M19 entered service in 1944 with the US Army, but did not see combat in WW2. It was used in the Korean War, mostly in against ground targets, but was retired in 1953 and replaced by the M42. It served in the

MEDIUM AAA UNITS

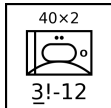
Netherlands Army from 1951 to 1978.⁵⁴



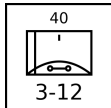
M34 Section: The M34 Gun Motor Carriage has a Bofors 40 mm L/60 cannon in an open mount on a modified M15 half-track. Many M15s were converted into M34s as supplies of 37-mm shells for the M1 cannon became scarce. Unlike the M15, the gun crew have no armored protection. A section has two vehicles. The M34 entered service with the US Army in 1951, saw combat in the Korean War, mainly in a ground-support role, and was likely retired in 1953.⁵⁵



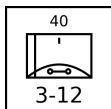
Bofors L/70 Battery: The Bofors L/70 has a single 40 mm gun in an open mount on a towed carriage. A battery has six gun mounts. Between the 1930s and the end of WW2, aircraft speed increased substantially, and so the engagement time of the Bofors L/60 became inadequate. The new L/70 fired a lighter shell at a higher velocity and achieved a significant improvement in range and engagement time. The basic fire-control system is optical, but various external fire-control radars can be used with it, including the Super Fledermaus and Flycatcher systems. NATO forces and others adopted it widely starting in 1952. Israel used batteries with the Super Fledermaus FCR for point-defense in the 1967 war.⁵⁶



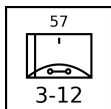
M42 Section: The M42 Duster has two Bofors 40 mm L/60 guns in a partially armored mount on a tracked chassis based on that of the M41 light tank. A section has two vehicles. The fire-control system is entirely optical. The M42 entered service in 1953 with the US Army and US Marine Corps, replacing the M19, saw combat in the Vietnam War against ground targets, and served at least with reserve units until 1988. It was also used by Austria, Greece, Japan, Jordan, Lebanon, Pakistan, Taiwan, Thailand, Tunisia, Turkey, Venezuela, Vietnam (South and North), and West Germany.⁵⁷



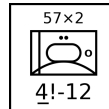
Bofors L/70 BOFI: The BOFI version of the Bofors L/70 adds modern sights, a laser rangefinder, and proximity-fuzed shells. A battery has six gun mounts. The BOFI is used by the armed forces of Brazil.⁵⁸



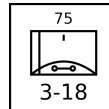
Bofors L/70 BOFI-R Battery: The BOFI-R version of the Bofors L/70 also adds an integrated fire-control radar to the BOFI version. A battery has six gun mounts.⁵⁹



S-60 Battery: The S-60 is a Soviet single 57 mm gun on an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but the gun is often used with the SON-9 (Fire Can) radar. The S-60 entered service in 1950 with Soviet forces, replacing the 61-K 73 mm gun. It was exported to many Soviet allies and saw combat with Arab forces in the 1967 and 1973 Arab-Israeli War, with communist forces in the Vietnam War, and with Iraqi forces in the Iran-Iraq War and the Gulf War. However, it was not used by communist forces in the Korean War. When paired with the SON-9 (Fire Can) radar, it was considered one of the more dangerous AAA systems faced by US aircraft in Vietnam.⁶⁰

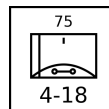


ZSU-57-2 Section: The ZSU-57-2 has two 57 mm cannons in a lightly armored but open-topped turret on an armored and tracked hull. A section has two vehicles. The cannons are similar to the S-60 cannons and the fire-control system is optical. The ZSU-57-2 entered service in the Soviet Army in 1955 and replaced the BTR-40A and BTR-152A self-propelled 14.5 mm AA guns in the AA batteries of tank regiments. It was in turn replaced by the ZSU-23-4 from 1965. The ZSU-57-2 also served widely with Soviet allies and saw combat with Egypt and Syria in the 1967 and 1973 Wars, with Syria in the 1982 Lebanon War, with North Vietnam in the 1972 Easter Offensive and the 1975 Spring Offensive, and with various factions in the Yugoslav Wars in the 1990s.⁶¹

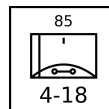


M51 Battery: The M51 Skysweeper has a single 75 mm gun in an open mount in a towed carriage. A battery has four mounts. The mount incorporates the T38 radar fire-control system and the gun has a relatively high rate of fire of 45 rounds per minute. Proximity-fuzed shells are available. The M51 entered service with the US Army in 1951 and was used for defense of strategic locations in the USA and in Europe, but was retired in 1959. It also served with Greece, Japan, and Turkey, with the last examples being retired until the mid-1970s.⁶²

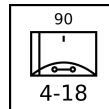
HEAVY AAA UNITS



Flab Kan 38 Battery: The Swiss Flab Kan 38 (Fliegerabwehr Kanone 38) has a single 75 mm L/49 gun in an open mount on a towed carriage. It is a licensed copy of the French Canon de 75 contre aéronefs modèle 1939 Schneider. A battery has six gun mounts. Its fire control system is optical, but in Swiss service from 1958 it was paired with the Mk.7 Blue Cedar radar. The Flab Kan 38 entered service with the Swiss Air Force in 1939 and was retired in 1967.⁶³

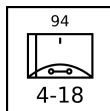


KS-12 Battery: The KS-12 (M1939 52-K) and the very similar KS-12A (M1944 KS-1) have a single 85 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The rate of fire of the KS-12 is 10 rounds per minute. Its fire-control system is optical, but the guns are often paired with the SON-9 (Fire Can) radar. The KS-12 entered service with Soviet forces in 1939. After WW2, it began to be replaced in Soviet service by the KS-19 100 mm and KS-30 130 mm guns, but was widely used by Soviet allies and saw combat with communist forces in the Korean War and the Vietnam War.⁶⁴

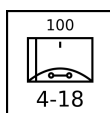


90 mm M2 Battery: The 90 mm M2 has a single 90 mm gun on a towed carriage. A battery has four guns. The fire control system of the M2 is optical, but an SCR-584 FCR-A normally controls each battery. The mechanical loading system permits a rate of fire of 24 rounds per minute. Proximity-fuzed shells are available. The M2 entered service with the US Army in 1943, following on from the similar M1 in 1940, and served as the main US heavy AA gun in WW2 and the

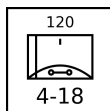
Korean War. It was retired in the mid 1950s. It also served with Canadian (until 1960), French, and Turkish forces.⁶⁵



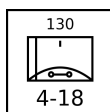
QF 3.7-inch Battery: The QF 3.7-inch has a single 94 mm gun on a towed carriage, but typically is used from a prepared firing site. The fire control system is optical, but an SCR-584 FCR-A normally controls each battery. In the final Mk VI version, the mechanical loading system permits a rate of fire of 20 rounds per minute. Proximity-fuzed shells are available. The QF 3.7-inch entered service with the British Army in 1937, served as its principal heavy AA gun during WW2 and beyond, and was withdrawn in 1957. It also served with Australia, Belgium, Canada, Sri Lanka, Cyprus, India, Ireland, Israel, Malta, Nepal, New Zealand, Pakistan, South Africa, Yugoslavia, and Portugal.⁶⁶



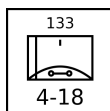
KS-19 Battery: The KS-19 has a single 100 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system of the KS-19 is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. The KS-19 entered service in 1948 with Soviet forces as a replacement for the KS-12 85 mm gun. It also served with many Soviet allies and saw combat with communist forces in the Korean War and Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.⁶⁷



120 mm M1 Battery: The M1 has a single 120 mm gun in an open mount on a towed carriage, but typically is used from a prepared firing site. A battery has four guns. The fire control system of the M1 is optical, but an SCR-584 FCR-A normally controls each battery. The rate of fire is 12 rounds per minute. Proximity-fuzed shells are available. The M1 entered service with the US Army in 1944, and though deployed to the Pacific Theater did not engage on enemy aircraft. After the war, it was widely deployed to defend of strategic targets against Soviet Tu-4 bombers, but was withdrawn by 1960. It also served with Canadian forces.⁶⁸



KS-30 Battery: The KS-30 has a single 120 mm gun in an open mount on a towed carriage. A battery has six gun mounts. Development of the KS-30 started after WW2, and it has features adopted from captured German guns as well as British QF 3.7-inch guns. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. The rate of fire is 12 rounds per minute. The KS-30 entered service in 1955 with Soviet forces, was used to defend strategic sites, and served until 1962. It was not widely exported, but also served with the North Vietnamese forces in the Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.⁶⁹



QF 5.25-inch Battery: The QF 5.25-inch has a single 133 mm gun in a fixed, fortified turret. It is a dual-purpose gun, used both against aircraft and for coastal defense, adapted from a naval gun. A battery has four guns. The fire control system is optical, but

an SCR-584 or Blue Cedar FCR-A normally controls each battery. The hydraulic loading system permits a rate of fire of 10 rounds per minute. The QF 5.25-inch entered service with the British Army in 1942 and was installed in and around London. Late in WW2, seven were installed in Australia and three in New Guinea. Eleven were installed in Gibraltar starting in 1953 and were manned until the early 1980s.⁷⁰

AAA UNITS TABLE

Type	Gun	Country	Year	Size ^a	Defense Strength	Sighting Range	Mobility	VPs		AAA Class	Altitude	Range			Hit			Damage Rating FCR/LRF	Night IR Sights	SAM
								3D/2D/D				S	M	L	S	M	L			
M2	12.7 mm	US	1933	Platoon	3	12	U	3/2/1	L	5	2	3	4	3	2	1	1	—	—	—
DShK-38	12.7 mm	SU	1938	Platoon	3	12	U	3/2/1	L	5	2	3	4	3	2	1	1	—	—	—
M16	12.7 mm × 4	US	1944	Section	3!	12	U	6/3	L	5	2	3	4	4	3	2	2	—	—	—
M17	12.7 mm × 4	US	1944	Section	3!	12	U	6/3	L	5	2	3	4	4	3	2	2	—	—	—
M55	12.7 mm × 4	US	1945	Platoon	3	12	T	4/3/1	L	5	2	3	4	5	4	3	2	—	—	—
Vz.53	12.7 mm × 4	CS	1953	Platoon	3	12	T	4/3/1	L	5	2	3	4	5	4	3	2	—	—	—
BTR-152A Vz.53	12.7 mm × 4	CS	1955	Section	3!	12	G/P	6/3	L	5	2	3	4	4	3	2	2	—	—	—
Mobile M45	12.7 mm × 4	US	1967	Section	2	12	G/P	4/2	L	5	2	3	4	4	3	2	2	—	—	—
ZPU-1	14.5 mm	SU	1949	Battery	3	12	T	3/2/1	L	6	2	3	5	3	2	1	1	—	—	—
ZPU-2	14.5 mm × 2	SU	1949	Battery	3	12	T	4/3/1	L	6	2	3	5	4	3	2	1	—	—	—
ZPU-4	14.5 mm × 4	SU	1949	Battery	3	12	T	5/3/2	L	6	2	3	5	5	4	3	2	—	—	—
Mobile ZPU-1	14.5 mm	SU	1949	Section	2	12	G/P	2/1	L	6	2	3	5	2	1	0	1	—	—	—
Mobile ZPU-2	14.5 mm × 2	SU	1949	Section	2	12	G/P	3/2	L	6	2	3	5	3	2	1	1	—	—	—
Mobile ZPU-4	14.5 mm × 4	SU	1949	Section	2	12	G/P	4/2	L	6	2	3	5	4	3	2	2	—	—	—
BTR-40A	14.5 mm × 2	SU	1950	Section	3!	12	G/P	4/2	L	6	2	3	5	3	2	1	1	—	—	—
BTR-152A ZPU-2	14.5 mm × 2	SU	1951	Section	3!	12	G/P	4/2	L	6	2	3	5	3	2	1	1	—	—	—
BTR-152A ZPU-4	14.5 mm × 4	SU	1951	Section	3!	12	G/P	5/3	L	6	2	3	5	4	3	2	2	—	—	—
M167	20 mm × 6	US	1967	Platoon	3	12	T	7/5/2	L	8	2	4	6	6	5	4	3	R/HF	—	—
M163	20 mm × 6	US	1968	Section	4!	12	U	8/4	L	8	2	4	6	5	4	3	3	R/HF	—	—
TCM-20	20 mm × 2	IL	1970	Platoon	3	12	T	5/3/2	L	8	2	4	6	4	4	3	2	—	—	—
M3 TCM-20	20 mm × 2	IL	1970	Section	3!	12	U	6/3	L	8	2	4	6	3	3	2	2	—	—	—
Single Rh-202	20 mm	DE	1972	Battery	3	12	T	4/3/1	L	8	2	4	6	4	3	2	2	—	—	—
Twin Rh-202	20 mm × 2	DE	1972	Battery	3	12	T	5/3/2	L	8	2	4	6	5	4	3	3	—	—	—
Panhard M3 VDA	20 mm × 2	FR	1975	Section	4	12	U	7/4	L	8	2	4	6	4	4	3	2	R/UF	—	—
ZU-23	23 mm × 2	SU	1960	Battery	3	12	T	6/4/2	L	9	2	5	8	5	4	3	3	—	—	—
Mobile ZU-23	23 mm × 2	SU	1960	Section	2	12	G/P	5/3	L	9	2	5	8	4	3	2	3	—	—	—
ZSU-23-4	23 mm × 4	SU	1965	Section	4	12	U	9/5	L	9	2	5	8	5	4	3	3	W/VF	—	—
BTR-152A ZU-23	23 mm × 2	PS	1975	Section	3!	12	G/P	4/2	L	9	2	5	8	4	3	2	3	—	—	—
Sinai 23 Ayn-al-Saqr	23 mm × 2	EG	1992	Section	4	12	U	10/5	L	9	2	5	8	5	4	3	3	R/UF	—	Ayn-al-Saqr
Sinai 23 Stinger	23 mm × 2	EG	1992	Section	4	12	U	12/6	L	9	2	5	8	5	4	3	3	R/UF	—	FIM-92
LAV-AD	25 mm × 5	US	1997	Section	4	12	U	14/7	L	10	3	6	9	5	4	3	4	—	Y	FIM-92D
M53/59	30 mm × 2	CS	1959	Section	4!	12	G/P	7/4	L	11	4	7	9	3	2	1	4	—	—	—
AMX-13DCA	30 mm × 2	FR	1969	Section	4	12	U	10/5	L	11	4	7	9	5	4	3	3	W/VF	—	—
AMX-30SA	30 mm × 2	FR	1979	Section	4	12	U	10/5	L	11	4	7	9	5	4	3	3	W/VF	—	—
Tunguska	30 mm × 2	SU	1984	Section	4	12	U	20/10	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19
Tunguska-M	30 mm × 2	SU	1990	Section	4	12	U	22/11	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19

AAA UNITS TABLE

		Type	Gun	Country	Year	Size ^a	Defense Strength	Sighting Range	Mobility	VPs		AAA Class	Altitude	Range			Hit			Damage Rating	FCR/LRF	Night IR Sights	SAM			
										3D/2D/D				S	M	L	S	M	L							
Oerlikon GDF	35 mm × 2			CH	1963	Battery	3	12	T				7/5/2	M	12	4	8		10	4	3	2	4	—	—	—
Oerlikon GDF	35 mm × 2			CH	1963	Section	3	12	T				5/2	M	12	4	8		10	3	2	1	4	—	—	—
Gepard	35 mm × 2			DE	1976	Section	4	12	U				11/6	M	12	4	8		10	5	4	3	4	W/VF	—	—
61-K	37 mm			SU	1939	Battery	3	12	T				5/3/2	M	12	4	8		11	3	2	1	4	—	—	—
M15	37 mm + 12.7 mm × 2			US	1942	Section	3	12	U				4/2	M	12	2	4		8	3	2	1	4	—	—	—
Bofors L/60	40 mm			SE	1935	Battery	3	12	T				5/3/2	M	15	4	8		12	2	2	1	4	—	—	—
M19	40 mm × 2			US	1944	Section	3	12	U				6/3	M	15	4	8		12	2	2	1	4	—	—	—
M34	40 mm			US	1951	Section	2	12	U				3/2	M	15	2	4		8	1	1	1	4	—	—	—
Bofors L/70	40 mm			SE	1952	Battery	3	12	T				6/4/2	M	15	4	8		12	3	3	2	4	—	—	—
M42	40 mm × 2			US	1953	Section	3	12	U				6/3	M	15	4	8		12	2	2	1	4	—	—	—
Bofors L/70 BOFI	40 mm			SE		Battery	3	12	T				8/5/3	M	15	4	8		12	5	5	4	4	—	—	—
Bofors L/70 BOFI-R	40 mm			SE		Battery	3	12	T				9/6/3	M	15	4	8		12	6	6	5	4	W/VF	—	—
S-60	57 mm			SU	1950	Battery	3	12	T				8/5/3	M	18	5	10		15	2	2	1	5	—	—	—
ZSU-57-2	57 mm × 2			SU	1955	Section	4	12	U				6/3	M	18	5	10		15	2	1	1	5	—	—	—
M51	75 mm			US	1953	Battery	3	18	T				9/6/3	M	30	6	12		18	4	4	3	5	W/MF	—	—

AAA UNITS TABLE

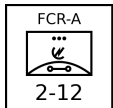
		Type	Gun	Country	Year	Size ^a	Defense Strength	Sighting Range	Mobility	VPs		AAA Class	Altitude	Range			Hit			Damage Rating	FCR/LRF	Night IR Sights	SAM						
										3D/2D/D				S	M	L	S	M	L										
Flab Kan 38	75 mm			CH	1938	Battery				4	18	T		9/6/3	H	24	6	12		18		2	1		0	5	—	—	—
KS-12	85 mm			SU	1939	Battery				4	18	T		9/6/3	H	27	6	12		18		2	1		0	6	—	—	—
90 mm M2	90 mm			US	1943	Battery				4	18	T		9/6/3	H	43	7	14		21		3	2		1	6	—	—	—
QF 3.7-Inch	94 mm			UK	1939	Battery				4	18	T		9/6/3	H	45	9	18		27		3	2		1	6	—	—	—
KS-19	100 mm			SU	1948	Battery				4	18	T		10/7/3	H	39	7	14		21		1	1		0	6	—	—	—
120 mm M1	120 mm			US	1944	Battery				4	18	T		9/6/3	H	57	11	22		33		2	2		1	7	—	—	—
KS-30	130 mm			SU	1955	Battery				4	18	T		11/7/4	H	64	12	24		36		1	1		0	7	—	—	—
QF 5.25-Inch	133 mm			UK	1942	Battery				4	18	T		11/7/4	H	46	12	24		36		2	2		1	7	—	—	—

FCR UNITS

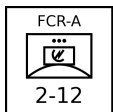
Most medium and heavy AAA batteries can be associated with an add-on fire-control radar (FCR), which improves their accuracy and allows them to engage unsighted targets at night and in adverse weather. These units are presented below with their counters.

The FCR Unit Table summarizes the properties of FCR units. As well as the basic properties common to all ground units, the table gives: the FCR class; the frequency band; and the modifier to the hit roll.⁷¹

FCR-A



Towed and Mobile FCR-A Platoon: FCR-As typically use 1940s technology. They can be towed or mobile.



SCR-584: The SCR-584 is an early US towed FCR-A based on the British 3 GHz magnetron. It replaced the much larger and less accurate SCR-268 and its basic technology was used in a series of successors. The SCR-584 entered service with US and British anti-aircraft artillery units in 1944 and was commonly used to control batteries of 90 mm M2, QF 3.7-inch, 120 mm M1, and QF 5.5-inch guns. It was also supplied to the Soviet Union and used to control batteries of 85 mm KS-12, and 100 mm KS-19 guns.⁷²

SCR-784: The SCR-784 towed FCR-A is a more compact version of the SCR-584. It entered service with the US Army in 1945, replacing the SCR-584 and controlling 90 mm M2 and 120 mm M1 guns.⁷³

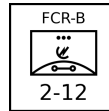
SON-4: The Soviet SON-4 (Whiff) towed FCR-A is based on the SCR-584 radars supplied by the US during WW2. It is commonly used with 85 mm KS-12 and 100 mm KS-19 batteries. It entered service in the Soviet Army in 1948 and started to be replaced by the SON-9 in 1955.⁷⁴

Blue Cedar: The Blue Cedar or "Radar, Anti-Aircraft No.3 Mk.7" towed FCR-A is a post-war development of the Mk.4 radar with post-war technology, resulting in a more compact and reliable system. It entered service with the British Army in 1952, controlling batteries of QF 3.7-inch and QF 5.25-inch guns, and served until these guns were retired in 1957. It also served with the Swiss Air Force from 1958, controlling 7.5 cm guns, until 1965, when the radars were replaced by Super Fledermaus systems and the guns by 35 mm Oerlikon GDFs. The radar was also used by the Netherlands, Sweden, Finland, and South Africa.⁷⁵

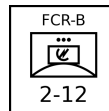
SON-9: The Soviet SON-9 (Fire Can) towed FCR-A was derived from the earlier SON-4, which in turn was derived from the US-supplied SCR-584. It is commonly used with 57 mm S-60, 85 mm KS-12, 100 mm KS-19, and 130 mm KS-30 batteries. The SON-9 entered service in 1955 with Soviet forces and was widely exported to Soviet allies. It has seen combat, including with North Vietnam in the Vietnam War, Egypt and Syria in the 1967 War, the War of Attrition, and the 1973 War.⁷⁶

SON-30: The Soviet SON-30 (Fire Wheel) towed FCR-B was derived from the earlier SON-9. It entered service in 1957 and was used to control batteries of 130 mm KS-30 guns.⁷⁷

FCR-B

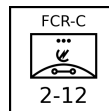


Towed and Mobile FCR-B Platoon: FCR-Bs typically use 1950s technology. They can be towed or mobile.

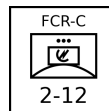


SON-50: The SON-50 (Flap Wheel) mobile FCR-B is new development and a contemporary of the RPK-2/1RL33 (Gun Dish) radar used in the ZSU-23-4. It has an integrated IFF system. It is used to control batteries of S-60 57 mm and KS-30 130 mm guns and entered service in the 1960s.⁷⁸

FCR-C



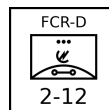
Towed and Mobile FCR-C Platoon: FCR-Cs typically use 1960s technology. They can be towed or mobile.



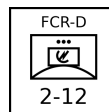
Super Fledermaus: The Contraves Super Fledermaus FCR-C can control a section of two Oerlikon GDF 35 mm guns; it cannot control a whole battery. It entered service in 1965 with the Swiss Air Force and also was used by the Argentine Air Force, Brazil, Iran, Israel, Japan, and South Africa. It was used by Israel in the 1967 Six Day War and afterward, controlling batteries of Bofors 40 mm L/70 guns in the defense of strategic sites. The Super Fledermaus also saw combat with the Argentine Air Force in the 1982 South Atlantic War controlling Oerlikon GDF guns in the defense of BAM Malvinas (Port Stanley Airport).⁷⁹

Flycatcher: The Signaal Flycatcher FCR-C can control up to six Bofors L/60 or L/70 guns or SAM launchers. Flycatcher entered service with the Royal Netherlands Air Force in 1979, serving for airfield defense and each controlling three modified Bofors 40 mm L/70 gun mounts. It subsequently served with the Indian Armed Forces from 1985, the Irish Defense Forces from 2003, the Royal Netherlands Army from 1987, the Thai Armed Forces from 1981, Turkey, and Venezuela.⁸⁰

FCR-D



Towed and Mobile FCR-D Platoon: FCR-Ds typically use 1970s technology. They can be towed or mobile.



Skyguard: The Contraves Skyguard FCR-D can control two Oerlikon GDF 35 mm guns and additionally a SAM launcher with AIM-9 Sparrow, RIM-9 Sea Sparrow, or Aspide missiles. Skyguard entered service in 1977 with the Swiss Air Force. It

FCR UNIT TABLE

Type	Size	Defense Strength	Sighting Range	Mobility	VPs	FCR		
						Class	Frequency	Modifier
					3D/2D/D			
Towed FCR-A	Platoon	2	12	G/P	6/4/2	A	LF	-1
Mobile FCR-A	Platoon	2	12	T	6/4/2	A	LF	-1
Towed FCR-B	Platoon	2	12	G/P	6/4/2	B	MF	-2
Mobile FCR-B	Platoon	2	12	G/P	6/4/2	B	MF	-2
Towed FCR-C	Platoon	2	12	G/P	7/5/2	C	VF	-2
Mobile FCR-C	Platoon	2	12	T	7/5/2	C	VF	-2
Towed FCR-D	Platoon	2	12	G/P	8/5/3	D	MW	-3
Mobile FCR-D	Platoon	2	12	T	8/5/3	D	MW	-3

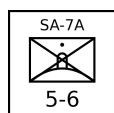
was subsequently very widely exported and used by the Argentine Army, Austria, Bahrain, Bangladesh, Brazil, Canada, Chile, China, Cyprus, Egypt, Greece, Indonesia, Iran, Italy, South Korea, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, Spain, Taiwan, and the UK. It saw combat with the Argentine Army in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield) controlling Oerlikon GDF 35 mm guns.⁸¹

INFANTRY SAM UNITS

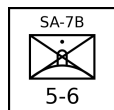
Infantry SAM units are squads of infantry specialized in the destruction of attacking aircraft with portable SAMs. These units are described below with their counters.

The Infantry SAM Unit Table summarizes the properties of infantry SAM units. In addition to the basic properties shared by all ground units, it gives: the number of ready, volley, and reload missiles; the multi-target capability; whether the unit has quick-reaction capability; whether the unit has IFF capability; whether the unit has night IR sights; and the optical lock-on number and range. Each squad has one launcher and, unless otherwise specified, one ready missile and two reload missiles. Infantry SAM squads may be attached to infantry, infantry weapons, infantry mortar, or infantry HQ platoons.

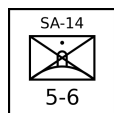
The Infantry SAM Table summarizes the properties of the actual missiles: guidance mode; launch roll; turn rate; flight time; visibility; ECCM, chaff, and flare ratings; whether the missile is active homing (AH) or has a home-on-jam (HOJ) capability; boost phase in FPs; base speed and sustainer duration in game turns; minimum altitude (and modifier for targets at T level); hit rolls for direct and proximity hits; and damage ratings for direct and proximity hits. Most infantry SAMs do not have true proximity fuzes, and so proximity hits in the game correspond to glancing direct hits in reality.



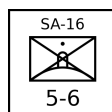
SA-7A Squad: The 9K32 Strela-2 (SA-7A Grail) was the first Soviet portable SAM. It has a rear-aspect, uncooled IR seeker. It entered service with Soviet forces in 1968, but quickly was replaced by the improved Strela-2M (SA-7B).⁸²



SA-7B Squad: The 9K32M Strela-2M (SA-7B Grail) is similar to the Strela-2 (SA-7A Grail), but uses a rear-aspect, electrically-cooled IR seeker. In Warsaw Pact armies, the Strela-2M was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-2M entered service with Soviet forces in 1970, replacing the Strela-2. It has seen combat in almost every conflict since 1970 in which the Soviet Union or its successor states have been participants or supporters. In particular, the Strela-2M was used heavily by North Vietnamese forces in the 1973 Easter Offensive, by Arab forces in the 1973 Yom Kippur War, by Angolan forces in the South African Border War, by the Afghan Mujahedeen in the Soviet-Afghan War (having been supplied by the CIA), by Iran in the Iran-Iraq War in the 1980s, and by Serbian and Serbian-supported forces in the Yugoslav Wars in the 1990s.⁸³

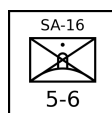


SA-14 Squad: The 9K34 Strela-3 (SA-14 Gremlin) is a development of the earlier Strela-2M. It has a rear-aspect, cryogen-cooled IR seeker. In Warsaw Pact armies, the Strela-3 was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-3 entered service with the Soviet Army in 1974. It has seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. In par-

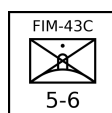


ticular, it was used by Iraqi forces in the 1991 Gulf War.⁸⁴

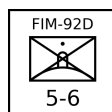
SA-16 Squad: The 9K310 Igla-1 (SA-16 Gimlet) uses as single-channel IR seeker similar to that in the SA-14 in a new missile body. The warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla-1 entered service with the Soviet Army in 1981. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters.⁸⁵



SA-18 Squad: The 9K38 Igla (SA-18 Grouse) uses a similar missile and launcher similar to the 9K310 Igla-1 (SA-16), but with a new a dual-channel IR seeker to reduce its susceptibility to flares. Again, the warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla entered service with the Soviet Army in 1983. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. It was used by Iraqi forces in the 1991 Gulf War, by the Peruvian Army in the 1995 Cenepa War, and by Serbian forces in the 1999 Kosovo War.⁸⁶



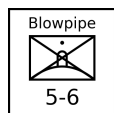
FIM-43C Squad: The FIM-43C Redeye Block III was the first US infantry SAM to reach production and enter service. It has a rear-aspect, cryogen-cooled IR seeker. The FIM-43C entered service with the US Army in 1968 and subsequently was used by Australia, Chad, Denmark, West Germany, Greece, Israel, Jordan, Saudi Arabia, Somalia, Sudan, Sweden, and Turkey. It was also supplied by the CIA to the Afghan Mujahedeen and the Nicaraguan Contras.⁸⁷



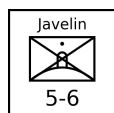
FIM-92A/B/C/D Squad: FIM-92A squad FIM-92B squad FIM-92C squad

The FIM-92A Stinger has an all-angle, second-generation, cryogen-cooled IR seeker. Its warhead weight 3 kg and was significantly more lethal than in earlier missiles. The FIM-92B Stinger POST adds a UV seeker to distinguish flares from aircraft. The FIM-92C/D Stinger RMP have software updates to further improve the performance against flares. All have an IFF interrogator to help avoid fratricide. The FIM-92A entered service with the US Army in 1981, the FIM-92B in 1986, the FIM-92C in 1989, and the FIM-92D in 1992. They were exported to dozens of US allies including the Afghan Mujahedeen and the Angolan UNITA. They saw service in the 1982 South Atlantic War, the Soviet-Afghan War, the Angolan Civil

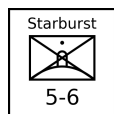
War, the Libyan Invasion of Chad, the Tajik Civil War, the Chechen War, the Sri Lankan Civil War, the Syrian Civil War, and the Russian Invasion of Ukraine.⁸⁸



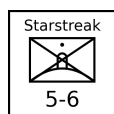
Blowpipe Squad: Blowpipe is an MCLOS optically-guided missile. While in theory this gave it an all-aspect capability, in practice guiding the missile to the target in combat proved to be very difficult. Blowpipe entered service with the British Army and Royal Marines in 1975. It was also used by the Argentine Army, Canadian Army, Chilean Army and Navy, Ecuadorean Army, Guatemalan Army, Portuguese Army, Malawian Army, Malaysian Army, Nigerian Army, Omani Army, Royal Thai Air Force and Army, and the UAE Army. It saw combat in the 1982 South Atlantic War with both sides and had a low success rate. In 1986, it was trialed by the Afghan Mujahedeen, but again was not considered to be effective. Finally, it was also used in the 1995 Cenepa War by Ecuador.⁸⁹



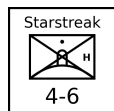
Javelin Squad: Given the poor performance of Blowpipe in the 1982 South Atlantic War, Javelin was quickly developed as an upgraded version of Blowpipe with SACLOS guidance in place of MCLOS guidance. Javelin replaced Blowpipe in the British Army and Royal Marines in 1984. It was also used by the Botswana Army, the Canadian Army, the Malaysian Army, the Peruvian Army, and the South Korean Army.⁹⁰



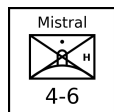
Starburst Squad: Starburst is Javelin with the radio-link replaced by a with laser link. It replaced Javelin in the British Army and Royal Marines in 1989 and was also used by the Canadian Army, the Malaysian Army, the Qatari Army, and the Royal Thai Army. Starburst served with the British Army in the 1991 Gulf War, but was not used in combat.⁹¹



Starstreak Squad: Starstreak is a high-speed laser-guided missile. As the missile approaches the target, it launches three independent explosive darts. The darts have contact fuzes, but not proximity fuzes. Starstreak replaced Javelin in the British Army and Royal Marines in 2000 and is also used by the South African Army and the Armed Forces of Ukraine. Starstreak served with the British Army in the 2003 Invasion of Iraq, but was not used in combat. It has seen combat in the Russian Invasion of Ukraine.⁹²

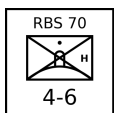


Starstreak LML Squad: The Starstreak Light Multiple Launcher (LML) has three ready-to-fire Starstreak missiles on a pedestal mount with a single sighting unit. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. Starstreak LML entered service with the British Army and Royal Marines in 2000. It is also used by Indonesia, Malaysia, and South Africa.⁹³

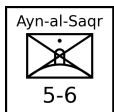


Mistral Squad: The Mistral is a French IR-homing missile on a pedestal mount. The missile has a considerably larger warhead than either the Stinger or Igla missiles, but this comes at a cost in portability. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The

Mistral entered service with the French Army in 1990. It was subsequently exported to many other countries.⁹⁴



RBS 70 Squad: The RBS 70 is a Swedish laser-guided missile on a pedestal mount. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The RBS 70 entered service with the Swedish Army in 1977. It was subsequently used by Argentina, Australia, Bahrain, Brazil, the Czech Republic, Iran, Ireland, Lithuania, Norway, Pakistan, Singapore, Thailand, Tunisia, the UAE, Ukraine, and Venezuela.⁹⁵



Ayn-al-Saqr Squad: The Ayn-al-Saqr ("Eye of the Hawk") missile is a licensed version of the 9M32M Strela-2m (SA-7B Grail) manufactured in Egypt. It equips one version of the Sinai 23 air-defense vehicle. It entered service with the Egyptian Army in 1984.⁹⁶

INFANTRY SAM UNIT TABLE

Name	Size	Year	Defense Strength	Sighting Range	Mobility	VPs		SAM	Missiles			Multi-Target	Quick Reaction	IFF	Night IR Sights	Lock-On		Other Names
						3D/2D/D			Ready	Volley	Reload					Optical	Range	
SA-7A	Squad	1968	5	6	U	-/-/3		SA-7A	1	1	2	-	Y	-	-	7	9	9K32 Strela-2 (Grail)
SA-7B	Squad	1972	5	6	U	-/-/4		SA-7B	1	1	2	-	Y	-	-	7	9	9K32M Strela-2M (Grail)
SA-14	Squad	1974	5	6	U	-/-/6		SA-14	1	1	2	-	Y	-	-	7	12	9K34 Strela-3 (Gremlin)
SA-16	Squad	1981	5	6	U	-/-/8		SA-16	1	1	2	-	Y	Y	-	7	12	9K310 Igla-1 (Gimlet)
SA-18	Squad	1983	5	6	U	-/-/10		SA-18	1	1	2	-	Y	Y	-	8	16	9K38 Igla (Grouse)
FIM-43C	Squad	1968	5	6	U	-/-/4		FIM-43C	1	1	2	-	Y	-	-	7	9	Redeye Block III
FIM-92A	Squad	1981	5	6	U	-/-/9		FIM-92A	1	1	2	-	Y	Y	-	8	12	Stinger
FIM-92B	Squad	1986	5	6	U	-/-/10		FIM-92B	1	1	2	-	Y	Y	-	8	12	Stinger POST
FIM-92C	Squad	1989	5	6	U	-/-/10		FIM-92C	1	1	2	-	Y	Y	-	8	12	Stinger RMP
FIM-92D	Squad	1992	5	6	U	-/-/10		FIM-92D	1	1	2	-	Y	Y	-	8	12	Stinger RMP
Blowpipe	Squad	1975	5	6	U	-/-/6		Blowpipe	1	1	2	-	Y	Y	-	7	9	
Javelin	Squad	1984	5	6	U	-/-/8		Javelin	1	1	2	-	Y	Y	-	7	9	Javelin GL
Starburst	Squad	1989	5	6	U	-/-/8		Starburst	1	1	2	-	Y	Y	-	7	9	Javelin S15
Starstreak	Squad	2000	5	6	U	-/-/12		Starstreak	1	1	2	-	Y	Y	-	7	12	
Starstreak LML	Squad	2000	4	6	U	-/-/14		Starstreak	3	1	0	-	Y	Y	-	7	12	
Mistral	Squad	1990	4	6	U	-/-/11		Mistral	1	1	2	-	Y	-	-	7	12	
RBS 70	Squad	1977	4	6	U	-/-/8		RBS 70	1	1	2	-	Y	Y	-	7	10	
Ayn-al-Saqr	Squad	1984	5	6	U	-/-/4		Ayn-al-Saqr	1	1	2	-	Y	-	-	7	12	

INFANTRY SAM TABLE

Name	Year	Guidance Mode	Launch Roll	Turn Rate	Flight Time	Visibility	ECCM	Chaff	Flare	AH	HOJ	Instant Arming	Boost Phase	Base Speed	Sustainer	Minimum Altitude	Hit		Damage		Other Names
																	Direct	Proximity	Direct	Proximity	
SA-7A	1968	I	6	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	6	4	2	9M32 Strela-2 (Grail)
SA-7B	1972	I	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	6	4	2	9M32M Strela-2M (Grail)
SA-14	1974	M	7	BT	1	6	-	-	4	-	-	Y	-	9	0	T	6	7	4	2	9M34 Strela-3 (Gremlin)
SA-16	1981	M	8	BT/2	1	6	-	-	4	-	-	Y	-	12	0	T	6	7	4*	2*	9K310 Igla-1 (Gimlet)
SA-18	1983	A	8	BT/2	1	6	-	-	2	-	-	Y	-	12	0	T	7	8	4*	2*	9K38 Igla (Grouse)
FIM-43C	1968	M	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	6	7	4	2	Redeye Block III
FIM-92A	1981	A	8	BT/2	1	2	-	-	3	-	-	Y	-	14	0	T	7	8	5	3	Stinger
FIM-92B	1986	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger POST
FIM-92C	1989	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger RMP
FIM-92D	1992	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger RMP
Blowpipe	1975	OG	8	HT	1	6	-	3	0	-	-	-	-	8	0	T	3	7	6	5	
Javelin	1984	OG	8	ET	1	5	-	2	0	-	-	Y	-	10	0	T	5	8	6	5	Javelin GL
Starburst	1989	LG	8	ET	1	5	-	1	0	-	-	Y	-	10	0	T	5	8	6	5	Javelin S15
Starstreak	2000	LG	8	ET	1	5	-	1	0	-	-	Y	-	24	0	T	7	9	6	4	
Mistral	1990	A	8	ET	1	5	-	-	2	-	-	Y	-	18	0	T	7	9	6	3	
RBS 70	1977	LG	8	ET	1	6	-	2	0	-	-	-	-	12	0	T	5	8	6	3	
Ayn-al-Saqr	1984	I	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	7	4	3	

EXAMPLE ORDERS OF BATTLE

This appendix contains a number of example orders of battle and their correspondence in game units.

The orders of battle for battalions are presented with the headquarters unit in the top row, next the organic companies, one per row, then any battalion-level assets on the next row or rows, and finally any attached units on the last line. Within each row, infantry or towed units appear on the left and their transport units on the right.

Unit Identifiers: The unit identifiers for line platoons are the platoon number, followed by the company letter or number, followed by the battalion number. If only one battalion is used, the battalion number may be omitted. Transport platoons will have a “T” prefix. For example, “1B3” means the first platoon of B company of the third battalion, and “T1B3” is its transport platoon. Companies normally have letters in Western armies and numbers in Warsaw Pact armies. Additionally, AD is used for air defense, AR for artillery, AT for anti-tank, R for reconnaissance, and S for scout, support, or security. The unit identifiers for battalion HQ units are HQ followed by the battalion number.

Unit Terminology: Within the game, units are referred to as a brigades, battalions, companies/batteries, and platoons according to their size. However, some forces use other terms, but not always with the same correspondence. The Unit Terminology Table gives examples of some of these uses.

UNIT TERMINOLOGY TABLE

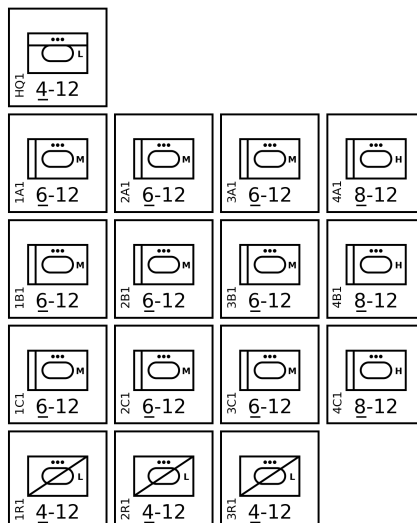
Force	Type	Brigade	Battalion	Company or Battery	Platoon
British Army	Armor	Brigade	<i>Regiment</i>	<i>Squadron</i>	<i>Troop</i>
	Artillery	—	<i>Regiment</i>	Battery	—
Soviet Army	Other	<i>Regiment</i>	Battalion	Company	Platoon
	Artillery	<i>Regiment</i>	Battalion	Battery	—
US Army	Cavalry	<i>Regiment</i>	<i>Squadron</i>	<i>Troop</i>	Platoon
	Pre-1965 Infantry	<i>Regiment</i>	Battalion	Company	Platoon
	Pre-1965 Armored	<i>Combat Command</i>	Battalion	Company	Platoon
USMC	Infantry	<i>Regiment</i>	Battalion	Company	Platoon

BRITISH ARMY (EARLY 1960S)

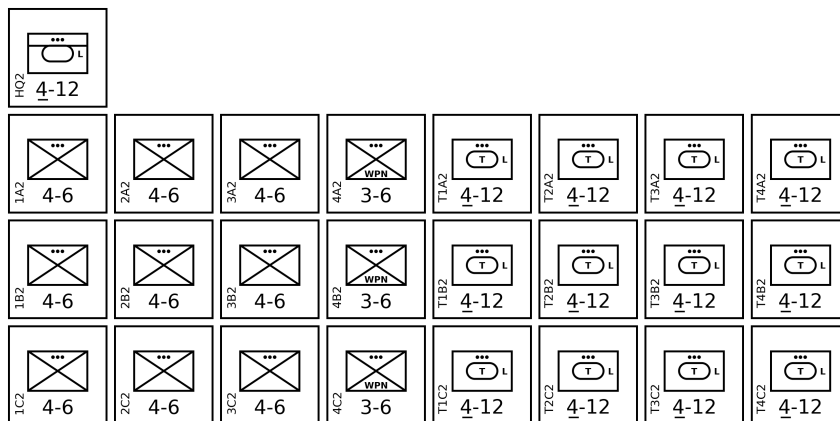
The British Army in the early 1960s was in transition. It operated both medium Centurion tanks (upgraded with the L7 105-mm gun) and heavy Conqueror tanks, and both mechanized and motorized infantry.⁹⁷

Armoured Regiment: This regiment has three squadrons A/B/C each with three tank troops with Centurion medium tanks with 105-mm guns and one tank troop with Conqueror heavy tanks with 120-mm guns. The Conqueror was originally intended to counter the Soviet IS-3 heavy tank, but after the Centurion was upgraded from the 84-mm 20-pdr gun to the 105-mm gun this need diminished. The reconnaissance squadron R has three light reconnaissance troops with Ferret armored cars.

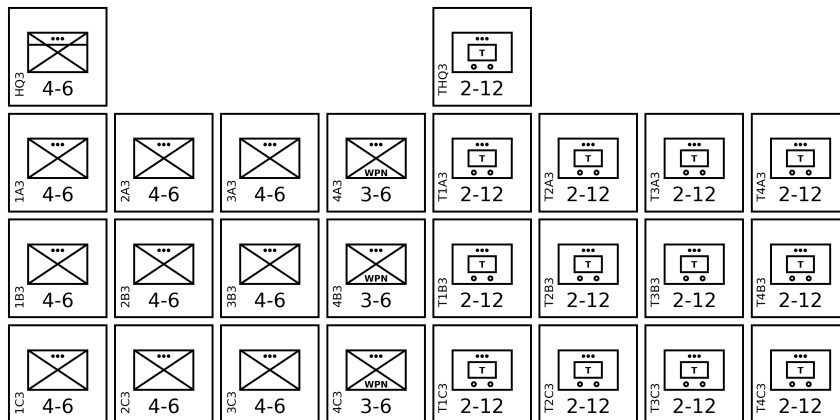
Note that in armored formations in the British Army, the terms troop, squadron, and regiment are used in place of platoon, company, and battalion.



APC Infantry Battalion: This battalion has three companies A/B/C each with three infantry platoons, one infantry weapons platoon, and corresponding light APC platoons with Saracen light APCs to provide mobility and protection. The organization parallels that of an ordinary infantry battalion, but with Saracens instead of trucks.

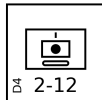
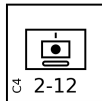
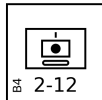
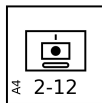
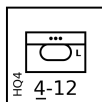


Infantry Battalion: This battalion has three companies A/B/C each with three infantry platoons, one infantry weapons platoon, with corresponding truck platoons to provide mobility.



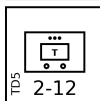
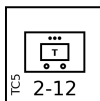
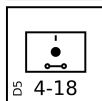
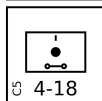
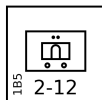
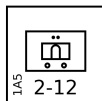
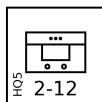
Field Artillery Regiment: This regiment has four batteries A/B/C/D each with six Cardinal 155-mm self-propelled guns.

Note that in artillery formations in the British Army, the term regiment is used in place of battalion.



Heavy Artillery Regiment: This regiment has four batteries A/B/C/D, of which A/B batteries each have two Honest John mobile missile launchers (and are each represented by section) and C/D batteries each have four towed 8-inch M1 guns with trucks.

Note that in artillery formations in the British Army, the term regiment is used in place of battalion.

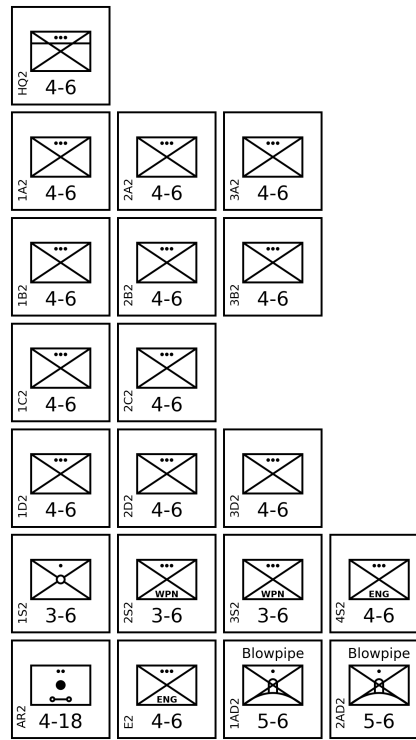


BATTLE OF GOOSE GREEN

The 1982 Battle of Goose Green in the South Atlantic War was fought between 2 Para Battalion of the British Army and 12 Infantry Regiment of the Argentine Army, both with attached elements.⁹⁸

2 Para Battalion: 2 Para consisted of: three line companies A/B/D each with three platoons of infantry; one patrol company C with two infantry platoons; and a support company S with one infantry mortar squad (as only two of the eight mortars in the mortar platoon were available), one infantry weapons platoon of sustained-fire GPMGs, one infantry weapons platoon with MILAN portable AT-GMs, and one infantry engineers platoon.

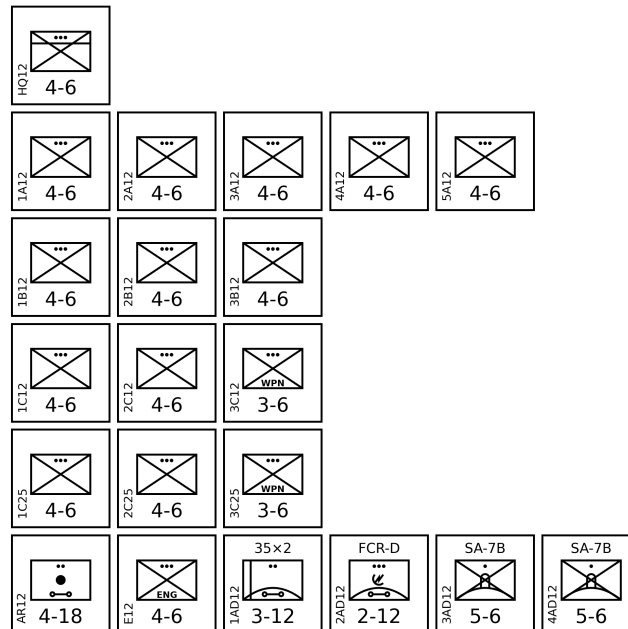
The attached units were: a towed artillery section of three 105-mm Light Guns from Alma Battery of 29 Commando Regiment, Royal Artillery; an infantry engineers platoon of 59 Independent Commando Squadron, Royal Engineers; and two squads of Blowpipes from 43 Air Defence Battery, Royal Artillery.



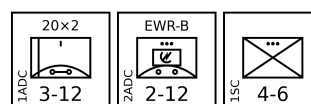
Task Force Mercedes: The Argentine Army elements at Goose Green were called *Fuerza de Taras Mercedes* (Task Force Mercedes).

The main body of the force was *Regimiento de Infantería 12* (12th Infantry Regiment), with three line companies A/B/C, and reinforced with C company of 25th Infantry Regiment.

The attached units were: a towed artillery section with three 105-mm pack howitzers from A Battery of 4th Airborne Artillery Regiment; an infantry engineers platoon from the 9th Engineering Company; a section with two Oerlikon GDF 35-mm AA guns with a Skyguard FCR from B Battery of 601st Anti-Aircraft Artillery Group; and a pair of SA-7B squads.



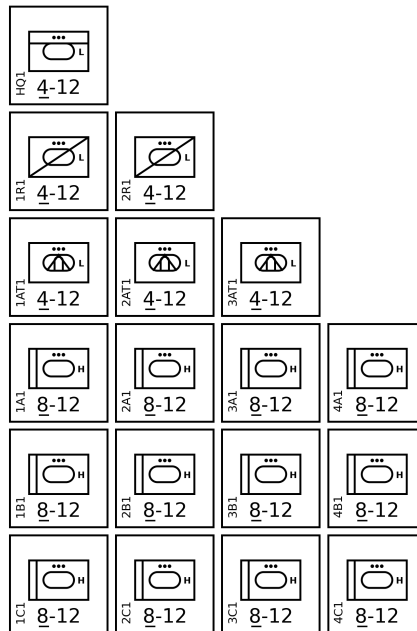
Military Airbase Condor: The Argentine Air Force named the airstrip at Goose Green as *Base Aérea Militar Cóndor* (Condor Military Air Base). The Air Force elements at the base included a battery of six 20-mm Rh-202 AA guns with an Elta EWR and a security platoon.



BRITISH ARMY (MID 1980S)

The British Army in the mid 1980s.⁹⁹

Armoured Regiment: This regiment [battalion] has three squadrons [companies] A/B/C each consisting of four tank troops [platoons] each with fourteen Chieftain heavy tanks. The reconnaissance (R) squadron [company] has two troops [platoons] each equipped with four Scorpion light tanks. The anti-tank squadron [company] has three troops [platoons] each equipped with three FV438s with the Swingfire missile.



ADDITIONAL RULES

RULE 1: GROUND UNITS

A ground *unit* is a combat or logistics formation of infantry, armor, transports, AAA, SAM launchers, or radars. This section discusses them in general.¹⁰⁰

A. Type and Size: A ground unit will have a type, for example, “heavy tank” or “ZSU-23-4”, and a size, which will be one of squad, section, platoon, or battery. Typically, a squad has five to ten soldiers and/or one vehicle, gun, or launcher; a section has ten to twenty soldiers and/or two vehicles, guns, or launchers; and a platoon or battery has thirty to fifty soldiers and/or three to six vehicles, guns, or launchers. The term “platoon” is more commonly used for infantry and tank units and “battery” for artillery and air-defense units, but within the game they are equivalent.

Companies, battalions, and other larger units above platoons or batteries are represented by multiple units. For example, an infantry company might be represented by three infantry platoon units.

All ground-combat and transport platoons are also available as sections and squads and all AAA platoons and batteries are also available as sections. A section or squad is identical in every way to a platoon or battery that has taken 1D or 2D damage, respectively, except that no VPs are awarded for this notional damage and non-AAA squads and sections may not engage in barrage fire.¹⁰¹

In the game, ground units are uniformly referred to as squads, sections, platoons, and batteries. However, other terms are used for these units even within English-speaking armies. For example, in many Commonwealth armies a section is equivalent to a squad, a troop is equivalent to a platoon, and a squadron is equivalent to a company. In the US Armored Cavalry, a troop is equivalent to a company and a squadron is equivalent to a battalion.

B. Basic Properties: The type and size of a ground unit determines its basic properties: damage resilience, defense strength, target class, mobility, and sighting range.

The damage resilience is the amount of damage a ground unit can suffer before being killed. It is determined by the size of the unit: a squad has a damage resilience of 1D; a section has a damage resilience of 2D; and a platoon or battery has a damage resilience of 3D.¹⁰²

The numerical defense strength is used to calculate the odds for air-to-ground attacks and for ground-to-ground combat.

The target class is used to determine the appropriate attack strength of any weapons used in air-to-ground attacks. The class can be a hard, open, or soft according to the degree to which the unit enjoys armored protection. Many weapons have different attack strengths against targets of different classes. A hard target has armored protection from both direct (front, side, and rear) and top attacks. Hard targets are further described as light, medium, or heavy according to their degree of protection, and have corresponding defense strengths.¹⁰³ An open target has armored protection from direct attacks but not from top attacks. An open target is normally treated as a hard target, but against napalm, fire, FAE, and air-burst HE bombs, it is considered to be a soft target.¹⁰⁴ A soft target does not have armored protection their personnel and/or weapons.¹⁰⁵

C. Representation: The briefing documents show the graphical representation of ground units as counters.¹⁰⁶ The size of a unit is indicated by one, two, or three dots (for squads, sections, or platoons) or one vertical bar (for batteries) in an upper position. Hard targets are indicated by an underlined defense strength and open targets by an underline defense strength followed by an exclamation point.

Advanced Rule

D. Mobility: The mobility of a ground unit restricts the hexes it can occupy according to the terrain in and around those hexes. Infantry units, tracked armored units, wheeled armored units other than armored trucks, and tracked unarmored units have unrestricted (U) mobility. Trucks, armored trucks, and other wheeled unarmored units have good (G) or poor (P) mobility. All other units have towed (T) mobility.¹⁰⁷

A unit with unrestricted mobility may occupy: any urban hex; any clear hex; or any forest hex. A unit with good mobility may occupy: any urban hex; any clear hex; or any forest hex with a road or trail. A unit with poor mobility may occupy: any urban hex; any clear or forest hex with a road, trail, or runway; or any clear hex adjacent to an urban hex or another clear hex with a road, trail, or runway. A unit with towed mobility may occupy hexes according to the mobility of its transport.¹⁰⁸

When necessary, a scenario must specify whether trucks, armored trucks, and other wheeled unarmored units have good or poor mobility, according to the terrain and the capabilities of the vehicles represented, and assign transport units to towed units.¹⁰⁹ A scenario may modify the mobility classes of ground units and the associated restrictions.¹¹⁰

E. Composite Platoons: At the start of the game, an infantry, infantry weapons, infantry mortar, or infantry HQ platoon may attach one or two infantry squads to form a composite platoon. The attached squads are commonly infantry SAM or FAC squads. Once formed, a composite platoon may not be split into its constituent units.¹¹¹

An attached squad is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. Instead, it is considered to form part of the composite platoon. An attached squad retains its normal capabilities (for example, an attached SAM squad may engage aircraft and an attached FAC squad may sight and mark targets).

A composite platoon has the same damage resilience, defense strength, target class, mobility, and sighting range as its constituent platoon. If a composite platoon is sighted, no information is given on any attached squads. If a composite platoon is identified, any attached squads are also identified (for example, “infantry platoon with an attached FAC squad”). If a composite platoon is suppressed, each attached squad is suppressed too. If a composite platoon suffers damage, then for each D of damage, roll one die for each attached squad and on a 3– the attached squad is killed. If a composite platoon is killed, each attached squad is killed too. VPs are awarded for damage to each of the constituent units.¹¹²

F. Transport: An infantry or towed unit may be transported by a specific associated transporting IFV, APC, armored truck, truck, or tracked-carrier unit of at least the same size. When necessary,

a scenario will establish the association between units and their specific transports.¹¹³

A transported unit is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. It may not attack other ground units or aircraft, may not launch SAMs, may not use radar, and may not act as a FAC.

If a transporting unit is sighted, no information is given on the transported unit. If a transporting unit is identified, no information is given on a transported infantry unit, but any transported towed unit is also identified (for example, “truck platoon towing an artillery battery” or “truck platoon towing a ZPU-4 battery”). If a transporting unit suffers damage, then its transported unit suffers the same damage. VPs are awarded for damage to both the transporting and transported unit.

A unit may start the game either transported or not. If not transported, it may be placed according to either its mobility or that of its associated transport.¹¹⁴

An infantry unit in the same hex as its associated transport may declare that it is mounting the transport at the start of a Ground Unit Interaction Phase. Both units must not be suppressed, and the transport must not have suffered more damage than the infantry unit. At the end of the Ground Unit Interaction Phase five game turns later, the unit is considered to be transported, and its counter is removed from the map.

An infantry unit being transported may declare that it is dismounting at the start of a Ground Unit Interaction Phase. The transport unit must not be suppressed. At the end of the Ground Unit Interaction Phase two game turns later, the unit is considered to be no longer transported, and its counter is placed on the map in the same hex as its transport. If placing the counter would violate stacking requirements, the unit must abort the dismounting process.

During these processes, the infantry unit may not perform any other action and the transport may not move and may not carry out ground attacks, but may perform barrage fire if it has this capability. The infantry unit may abort either of these process at the start of any Ground Unit Interaction Phase. If either are suppressed during the process, the unit must abort the process in the next Ground Unit Interaction Phase.¹¹⁵

A towed unit may not mount or dismount during the game.

NOTES